

Internship Report — Day 9

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Internship Role: Cloud / Infrastructure Deployment

Technology Focus: Azure Resource Manager (ARM) Templates

Date: Day 9 of Internship

Objective of the Day

The objective of Day 9 was to gain hands-on experience with Azure Resource Manager (ARM) templates by observing, understanding, and deploying infrastructure through automation. The focus was on learning the deployment workflow, resource dependencies, and template structure.

Task 1 — Deploy ARM Template and Observe Resource Creation Flow

I was assigned an ARM template deployment that provisions the following resources:

- Storage Account
- Virtual Machine
- File Share

Key Learning Goals

- Understand the deployment procedure
- Observe the order in which resources are created
- Identify resource dependencies

Observations

1. **Resource Group Validation**
 - Deployment begins by validating the template and parameters.
 - Resource group acts as the container for all resources.
2. **Storage Account Deployment**
 - Storage account is created first because it may be required by other services (e.g., diagnostics, file share).
3. **File Share Creation**
 - File share is created inside the storage account.
 - This confirms dependency on the storage account.

4. Virtual Machine Deployment

- VM provisioning happens after networking and storage components are prepared.
- VM depends on:
 - Virtual network
 - Network interface
 - Public IP
 - Storage for OS disk or boot diagnostics

Resource Dependency Understanding

Resource	Depends On
File Share	Storage Account
Network Interface	Virtual Network + Subnet
Virtual Machine	Network Interface + Storage
Boot Diagnostics	Storage Account

Deployment Insight

ARM templates automatically manage dependency order using the `dependsOn` property or implicit references.

Task 2 — Study ARM Template Structure

The second task involved reading and analyzing the ARM template in detail to understand its structure and components.

ARM Template Core Structure

1. **Schema**
 - Defines template format version.
2. **Parameters**
 - Inputs provided during deployment.
 - Example: VM name, OS type, admin credentials.
3. **Variables**
 - Derived values to avoid repetition.
 - Used for naming conventions and configurations.
4. **Resources**
 - Actual infrastructure components to deploy.
5. **Outputs**
 - Values returned after deployment (e.g., public IP).

Key Understanding

- Templates follow a JSON format.
 - Parameters make templates reusable.
 - Variables simplify configuration logic.
 - Resource blocks define infrastructure.
 - Outputs provide deployment results.
-

Task 3 — Conditional OS Deployment with Web Server Installation

The third task was to deploy a virtual machine using an ARM template with **conditional OS selection**.

Deployment Requirements

The template should allow switching between:

Option 1 — Linux VM

- OS: Linux
- Web Server: Nginx
- Custom HTML page deployed via script
- Access via Public IP

Option 2 — Windows VM

- OS: Windows
- Web Server: IIS
- Custom HTML page deployed
- Access via Public IP

Implementation Concept

- Boolean or parameter-based OS selection.
- Conditional resource configuration.
- Custom script extension used for software installation.

Verification Process

After deployment:

1. Retrieve VM public IP.
2. Open browser.
3. Access hosted HTML page.
4. Confirm web server is running correctly.

Result

- Linux deployment successfully hosted webpage using Nginx.
 - Windows deployment successfully hosted webpage using IIS.
 - Conditional logic in ARM template worked correctly.
-

Skills and Knowledge Gained

- Understanding infrastructure as code (IaC).
 - Learning ARM template architecture.
 - Identifying resource dependencies.
 - Using conditional deployment logic.
 - Automating software installation with custom scripts.
 - Validating web service accessibility via public IP.
-

Conclusion

Day 9 provided practical exposure to automated infrastructure deployment using ARM templates. I gained hands-on understanding of how Azure provisions resources, manages dependencies, and enables flexible deployments through parameterization and conditional logic. This experience strengthened my knowledge of Infrastructure as Code and cloud automation practices.

```
MINGW64: c:/Users/Lenovo/Spektra-Systems-Intern-
--u, --[no-]upload-pack <path> checkout <branch> instead of the remote's HEAD
--[no-]depth <depth> path to git-upload-pack on the remote
--[no-]shallow-since <time> create a shallow clone on the remote
--[no-]shallow-exclude <ref> create a shallow clone since a specific time
--[no-]single-branch <ref> deepen history of shallow clone, excluding ref
--[no-]tags <ref> clone only one branch, HEAD or --branch
--[no-]tags <ref> don't clone any tags, and make later fetches not to follow them
--[no-]shallow-submodules <ref> opposite of --no-tags
--[no-]separate-git-dir <gitdir> any cloned submodules will be shallow
--[no-]ref-format <format> separate git dir from working tree
--c, --[no-]config <key=value> specify the reference format to use
--[no-]server-option <server-specific> set config inside the new repository
--4, --ipv4 <server-specific> option to transmit
--6, --ipv6 <server-specific> use IPv4 addresses only
--[no-]filter <args> use IPv6 addresses only
--[no-]also-filter-submodules <args> object filtering
--[no-]remote-submodules <args> apply partial clone filters to submodules
--[no-]sparse <args> any cloned submodules will use their remote-tracking branch
--[no-]bundle-uri <uri> initialize sparse-checkout file to include only files at root
a URI for downloading bundles before fetching from origin remote

Lenovo@DESKTOP-A3HTFSD MINGW64 ~ (master)
$ git clone ^[[200~
fatal: repository '^[200~' does not exist

Lenovo@DESKTOP-A3HTFSD MINGW64 ~ (master)
$ git clone https://github.com/syedfuzailabdu11a/Spektra-Systems-Intern-.git
Cloning into 'Spektra-Systems-Intern-'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), 1.05 MiB | 1.17 MiB/s, done.
Resolving deltas: 100% (1/1), done.
Lenovo@DESKTOP-A3HTFSD MINGW64 ~ (master)
$ cd s
```

azure portal login - Search x Microsoft.Template-20260212093: x +

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Home

Microsoft.Template-20260212093356 | Overview

Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.Template-20260212093356

Subscription : Vanguard Class Training

Resource group : syedfuzail2009-rg

Start time : 2/12/2026, 9:34:02 AM

Correlation ID : e3b042bc-b7e1-4578-813e-1879b0d20a62

> Deployment details

✓ Next steps

Go to resource

Add or remove favorites by pressing Ctrl+Shift+F

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azure portal login - Search x Microsoft.Template-20260212132604 x +

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Microsoft.Template-20260212132604 | Overview

Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.Template-20260212132604 Start time : 2/12/2026, 1:26:06 PM
Subscription : Vanguard Class Training Correlation ID : e78ebabc-da35-46e3-bf1e-bd...
Resource group : syedfuzail2009-rg

Deployment details

Resource	Type	Status	Operations
syedfuzailstora...	Storage account	Accepted	Operations
syedfuzail-vnet	Virtual network	OK	Operations
syedfuzail-nsg	Network security group	OK	Operations

Add or remove favorites by pressing Ctrl+Shift+F

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Microsoft.Template-20260212132604 | Overview

Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment succeeded

Deployment 'Microsoft.Template-20260212132604' to resource group 'syedfuzail2009-rg' was successful.

Pin to dashbo... Go to resource gr...

Your deployment is complete

Deployment name : Microsoft.Template-20260212132604 Start time : 2/12/2026, 1:26:09 PM
Subscription : Vanguard Class Training Correlation ID : e78ebabc-da35-46e3-bf1e-bd...
Resource group : syedfuzail2009-rg

Deployment details

Next steps

Go to resource group

Give feedback

Tell us about your experience with deployment

Add or remove favorites by pressing Ctrl+Shift+F

Cost management

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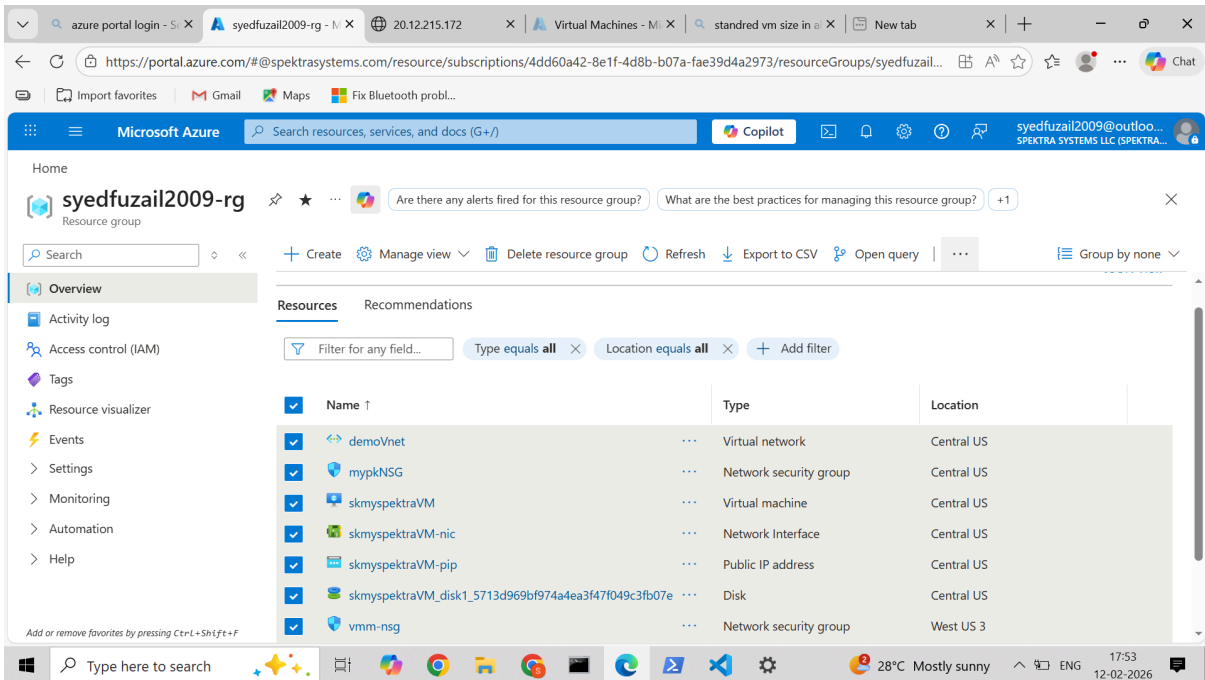
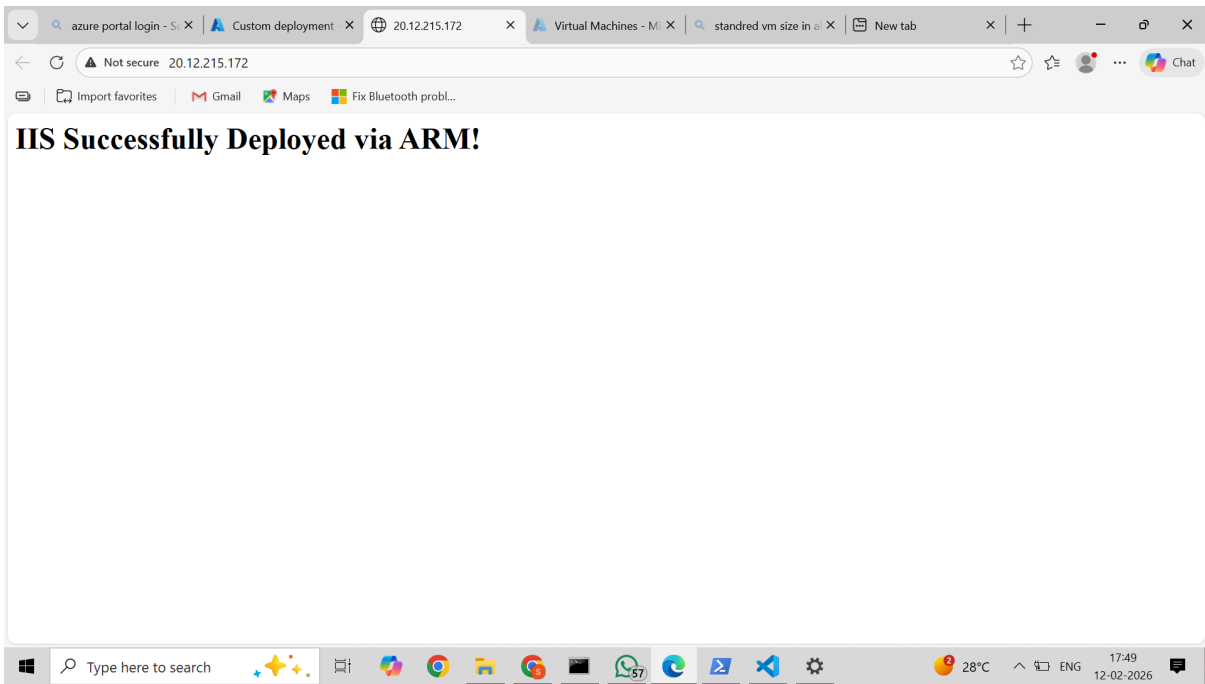
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Home

syedfuzail2009-rg

Resource group

Are there any alerts fired for this resource group? What are the best practices for managing this resource group? +1

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Group by none

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Monitoring
- Automation
- Help

Name	Type	Location
demoVnet	Virtual network	Central US
mypkNSG	Network security group	Central US
skmyspektraVM	Virtual machine	Central US
skmyspektraVM-nic	Network Interface	Central US
skmyspektraVM-pip	Public IP address	Central US
skmyspektraVM_disk1_5713d969b974a4ea3f47f049c3fb07e	Disk	Central US
vmm-nsg	Network security group	West US 3
vmm-vnet	Virtual network	West US 3

Showing 1 - 8 of 8. Display count: auto

Give feedback

Home > syedfuzail2009-rg

mypkNSG

Network security group

Search

Move Delete Refresh Give feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Essentials

Resource group (move)
[syedfuzail2009-rg](#)

Location
Central US

Subscription (move)
[Vanguard Class Training](#)

Subscription ID
4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

Tags (edit)
[Add tags](#)

Custom security rules
2 inbound, 0 outbound

Associated with
1 subnets, 0 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority ↑ Name ↑ Port ↑ Protocol ↑ Source ↑ Destination ↑ Action

Inbound Security Rules